Unit 6 Pre-Test

Quadratic Expressions & Equations

Standards assessed: NC.M1.A-APR.1 · A-APR.3 · A-SSE.3 · A-REI.4 · F-IF.8a

Name: _____ **Date:** _____ **Period:** _____

SCORE *3.0* **Basic Understanding**

NC.M1.A-APR.1 · NC.M1.A-APR.3 · NC.M1.A-SSE.3 · NC.M1.A-REI.4 Multiply and add polynomials, factor out a GCF, and solve simple quadratics.

• *I can multiply and add polynomials, factor out a common factor, and solve with square roots or zero products.*

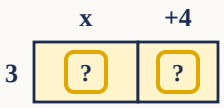
Answer each question. Use the picture to help you.

How to type your answers

power `x^2+5x+6` factored `(x+3)(x+4)` two answers `3, -5` ± `-2+sqrt(5)`

1. *Distribute*

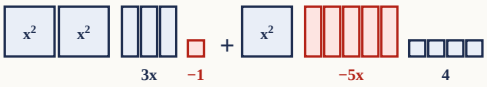
Multiply: 3(x + 4)



Answer: _____

2. *Add polynomials*

$(2x^2 + 3x - 1) + (x^2 - 5x + 4)$

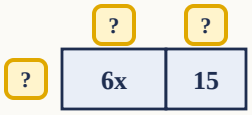


Answer: _____

3. *Factor (GCF)*

Factor out the greatest common factor:

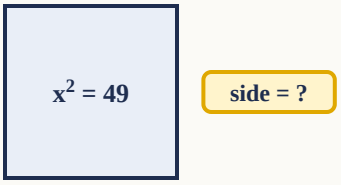
6x + 15



Answer: _____

4. *Square roots*

Solve: $x^2 = 49$



x = _____

5. *Zero product*

Solve: $(x - 3)(x + 5) = 0$

(x − 3)	(x + 5)
= 0 → x = ?	= 0 → x = ?

x = _____

6. *Multiply*

$(x + 3)(x + 2)$

	x	+3
x	?	?
+2	?	?

Answer: _____

SCORE 4.0 Target Content

NC.M1.A-APR.1 · NC.M1.A-APR.3 · NC.M1.A-SSE.3 · NC.M1.A-REI.4 Factor trinomials and differences of squares, and solve by factoring.

• *I can factor $x^2 + bx + c$ and $a^2 - b^2$, multiply binomials, and use factors to find zeros.*

Solve each problem. Show your work on paper.

7. *Factor*

Factor: $x^2 + 7x + 12$

multiply to **12**

? ?

7

add to

Answer: _____

8. *Factor*

Factor: $x^2 - 3x - 18$

multiply to **−18**

? ?

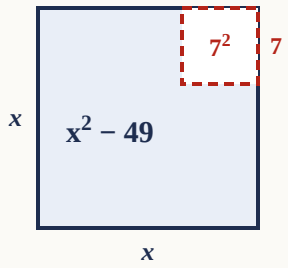
−3

add to

Answer: _____

9. *Difference of squares*

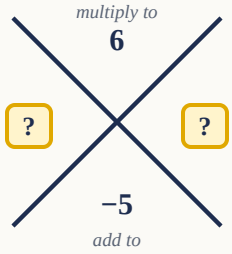
Factor: $x^2 - 49$



Answer: _____

10. *Solve by factoring*

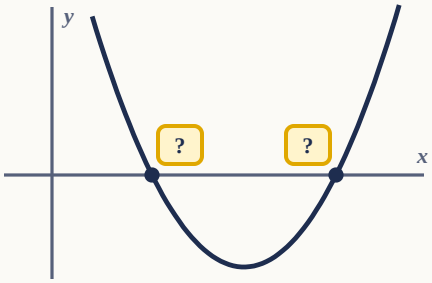
Solve: $x^2 - 5x + 6 = 0$



x = _____

11. *Zeros*

Find the zeros of f(x) = (x + 1)(x − 4).



x = _____

12. *Multiply*

(2x + 1)(x − 3)

	x	−3
2x	?	?
+1	?	?

Answer: _____

SCORE 5.0 Advanced Content

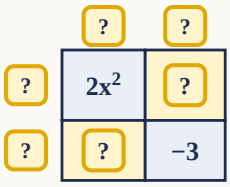
NC.M1.A-APR.3 · NC.M1.A-SSE.3 · NC.M1.A-REI.4 · NC.M1.F-IF.8a Factor with a ≠ 1, complete the square, and use the quadratic formula.

• *I can factor harder trinomials, complete the square, use the quadratic formula, and use forms to find key features.*

Solve each problem. Give every solution.

13. *Factor ($a \neq 1$)*

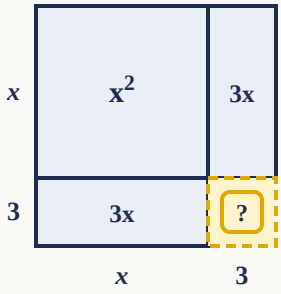
Factor: $2x^2 + 5x - 3$



Answer: _____

14. *Complete the square*

Solve by completing the square: $x^2 + 6x = 16$



x = _____

15. *Quadratic formula*

Solve: $x^2 + 4x - 1 = 0$. (Exact, or round to the hundredth.)

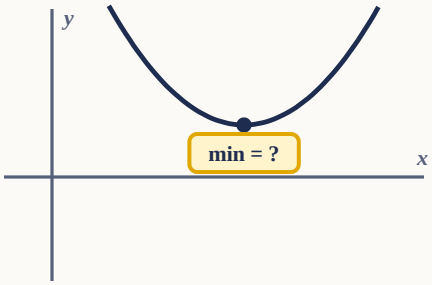
$$a = 1,\ b = 4,\ c = -1$$

$$x = \frac{-b \pm \sqrt{(b^2 - 4ac)}}{2a}$$

x = _____

16. *Vertex form*

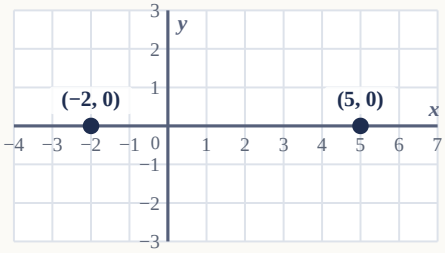
$x^2 - 6x + 11 = (x - 3)^2 + 2$. What is the **minimum value** of $y = x^2 - 6x + 11$?



min = _____

17. *Build from zeros*

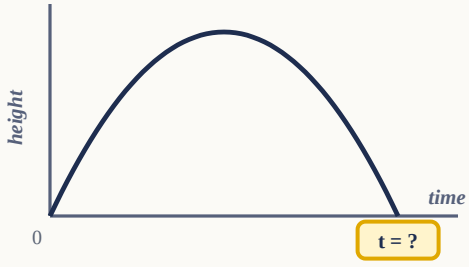
A quadratic has zeros $x = -2$ and $x = 5$, and $a = 1$. Write it in standard form.



y = _____

18. *Factored form*

A ball's height is $h(t) = -16t^2 + 32t = -16t(t - 2)$. After how many seconds does it land ($t > 0$)?



t = _____